NEXUS AI RESEARCH LAB | INTERNAL PROGRESS REVIEW | TOTAL DURATION: 7–10 MINUTES

STRATOS AI — Presentation & Demonstration Guide

Slide-by-Slide Outline, Speaking Script & Live Demonstration Walkthrough

Project: Strategic Team Reasoning & Tactical Operations | **Domain:** Multi-Agent Reinforcement Learning | **Date:** October 2026

Slide 1: Title & Introduction

1.0 MINUTE

SLIDE VISUALS:

- **TITLE:**
  STRATOS AI — Collective Intelligence. Coordinated Execution.
- **SUBTITLE:**
  Multi-Agent Reinforcement Learning and Inter-Agent Communication in Real-Time Strategic Combat.
- **HERO SCREENSHOT:**
  Top-down battlefield visualizer showing the 3 friendly units, crystal depots, and communication bus.

PRESENTER SCRIPT (READ OUT LOUD):

"Good morning/afternoon everyone. Today, I am proud to present **STRATOS AI**, an autonomous multi-agent reinforcement learning system that investigates a fundamental question in artificial intelligence:

'Does real-time communication between specialized AI agents measurably improve their coordination and tactical performance compared to agents operating independently?'

In real-time multi-agent environments—from autonomous drone swarms to robotic defense—individual agents often suffer from coordination breakdown. When agents can't talk to each other, they make localized, selfish decisions that lead to catastrophic failures. In STRATOS AI, we created a complete simulation testbed, designed a real-time tactical communication bus, and benchmarked the results across 150 independent trials."

Slide 2: Problem Statement & Research Methodology

1.5 MINUTES

SLIDE VISUALS:

- **CONTROLLED ABLATION GRAPHIC:**

  Physical observations (Channels 0–42) → Identical for both. Communication channels (Channels 43–48) → Active in Coordinated; Strictly Clamped to Zero in Independent.
- **3 OPPONENT ARCHETYPES:**

  Aggressive Rush (blitzkrieg), Defensive Turtle (base bunker), and Resource-Focused (economic expansion).

🎙️ **PRESENTER SCRIPT:**

"To make this a scientifically rigorous study, we avoided guesswork by applying strict ablation testing:

Both our **Coordinated squad** and our **Independent baseline squad** use the exact same neural policy weights, start on identical random seeds, and face identical unit placements.

The only difference: in Independent mode, the communication channels are strictly zeroed out. In Coordinated mode, the agents share real-time alerts like `ENEMY_DETECTED`, `DEFEND_BASE`, and `REQUEST_SUPPORT`.

We then pitted both configurations against three distinct opponent tactical behaviors: Aggressive Rushers, Defensive Turtlers, and Resource Harvesters."

Slide 3: System Architecture & Agent Roles

1.5 MINUTES

SLIDE VISUALS:

- **BATTLEFIELD MAP:**
  800 × 600 arena with Bases (1,000 HP), Crystal Depots, and Combat Range Circles.
- **SQUAD ROLES:**
  Commander (280 HP, macro priorities), Defense Guard (380 HP tank, HQ protection), Attack Striker (220 HP, 32 dmg scout).
- **NEURAL PIPELINE:**
  PettingZoo Parallel API → 49-Dim Observation → Actor-Critic Policy (PPO) → 7 Discrete Macro-Actions.

PRESENTER SCRIPT:

"The battlefield simulation is built according to PettingZoo parallel environment standards. Our friendly squad consists of three specialized units:
• The **Commander**, who sets strategic priorities,
• The **Defense Guard**, a high-armor tank that intercepts invaders,
• And the **Attack Striker**, a fast, high-damage unit specialized in long-range engagements.

At every step, each unit receives a normalized 49-dimensional observation vector and passes it through an Actor-Critic neural network trained using Proximal Policy Optimization (PPO). The policy outputs one of 7 macro-actions, including holding ground, perimeter defense, enemy engagement, or crystal harvesting."

Slide 4: Empirical Findings & Results (The Numbers)

2.0 MINUTES

SLIDE VISUALS: 150-EPISODE BENCHMARK COMPARISON TABLE (5 SEEDS)

Scenario / Opponent	Mode	Win Rate	Mean Reward	Avg Transmissions
AGGRESSIVE RUSH	Independent	**0 / 25 (0.0%)**	-408.89	0.0 msgs
AGGRESSIVE RUSH	Coordinated	**25 / 25 (100.0%)**	**+242.26**	819.0 msgs
DEFENSIVE TURTLE	Independent / Coord	100.0% vs 100.0%	+556.86	697.0 msgs (Coord)
RESOURCE FOCUSED	Independent / Coord	100.0% vs 100.0%	+408.82	503.0 msgs (Coord)

PRESENTER SCRIPT:

"Here are our key empirical findings across 150 benchmark episodes evaluated over 5 distinct random seeds:

1. **A Clear Win-Rate Advantage**: Coordinated agents achieved a perfect **100.0% win rate** across all scenarios, compared to **66.7%** for the independent baseline—a statistically verified **+33.3 percentage point advantage**.

2. **The Aggressive Rush Breakthrough**: When facing immediate aggressive base rushes, independent agents suffered a 100% squad wipeout loss (0.0% win rate). Why? Because individual agents wandered off toward crystal nodes and were picked off one by one.

In contrast, coordinated agents broadcasted emergency `DEFEND_BASE` radio directives immediately upon sighting hostiles. The entire squad retreated to form a tight defensive perimeter around the headquarters, completely neutralizing the enemy wave and winning 100% of matches.

3. **Communication Volume**: Coordinated agents exchanged an average of 673 tactical messages per match, verifying that our communication pipeline is active and driving real tactical decisions."

Slide 5: Live Demonstration

2.0 MINUTES

Terminal Command to Execute:
`.\.venv\Scripts\python.exe scripts/run_demo.py`

1. Point out the **Top HUD Bar** (Base HP, Crystal counts, Active Directive).
2. Point out the **Bottom Communication Ticker** (Live green radio feed).
3. **Press `[M]` Live**: Show that the radio goes silent (0 msgs) and units disperse. Press `[M]` again to resume coordination!
4. Point out the **End-of-Match Victory Card** with damage, survivors, and total messages.

PRESENTER SCRIPT:

"Now let's see the system live in action. *(Launch demo)*

On screen, you can see our battlefield. Notice the bottom ticker: this is our real-time tactical communication bus. As soon as the defense unit detects enemy units crossing midfield, it immediately sends high-priority radio alerts to the squad.

Notice what happens when I press the **[M]** key: we toggle into Independent Mode. The radio stream goes silent, communication channels are clamped to zero, and the units can no longer coordinate their defense.

When the match concludes, our tactical card shows the complete post-match telemetry—final headquarters health, surviving units, crystal mining totals, and total radio transmissions."

Slide 6: Research Analytics Dashboard & Conclusion

1.0 MINUTE

SLIDE VISUALS:

- Interactive Streamlit dashboard running in browser: `streamlit run dashboard/app.py`
- Summary: PettingZoo standard, PPO Actor-Critic, verified +33.3% communication win-rate advantage, 20/20 automated tests passing.

PRESENTER SCRIPT:

"In addition to the visual simulator, we built an interactive Streamlit research dashboard that lets researchers and evaluators inspect training dynamics, reward learning curves, and scenario-by-scenario distributions in real time.

All code is fully verified with a 20-test automated test suite passing at 100%.

In conclusion, STRATOS AI proves that structured, low-bandwidth communication channels prevent catastrophic failures and unlock significant, measurable performance gains in multi-agent reinforcement learning.

Thank you, and I would now be happy to take any questions."

Slide 7: Potential Evaluator Q&A Preparation

CHEAT SHEET

Q1: "How do you prove the agents are actually using the messages, rather than just reacting to seeing the enemy?"

Answer: "We performed controlled policy sensitivity audits where we fed the exact same physical coordinates into the network and only altered the 6 communication channels. We observed a measurable shift in action logits (Δ_{max} = 0.0516) and a +0.931 shift in critic state-value estimation. Furthermore, in our 150-episode multi-seed benchmark, independent agents facing identical enemy positions lost 100% of matches against aggressive rushers, while coordinated agents won 100%."

Q2: "Why did you use symbolic messages instead of continuous latent vectors (like CommNet)?"

Answer: "Symbolic messaging allows for full interpretability and debugging. Evaluators and human observers can read exactly what the AI is communicating (`DEFEND_BASE`, `REQUEST_SUPPORT`) in real time, making the system transparent and safe for human-in-the-loop applications."

Q3: "What is the purpose of the 15-step freshness filter?"

Answer: "Early in development, we found that without message expiration, emergency support requests stayed in agent inboxes forever, causing agents to stay in defensive fallback loops even after the enemy was already eliminated. The 15-step filter ensures that tactical directives expire once transient threats pass."